

March 5, 2015



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety Administration**

East Building, PHH – 30
1200 New Jersey Avenue, Southeast
Washington, D.C. 20590

DOT-SP 8965
(TWELFTH REVISION)

EXPIRATION DATE: February 28, 2019

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: PST Cylinders, LLC
Milwaukee, WI
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of non-DOT specification fiber reinforced plastic, hoop wrapped cylinder for use in the transportation of Division 2.1 and 2.2 hazardous materials described in paragraph 6 below. This special permit provides no relief from the Hazardous Materials Regulations other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.302a(a) in that DOT specification cylinders are required, except as specified herein.
5. BASIS: This special permit is based on the application of PST Cylinders, LLC dated January 29, 2015 in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/Division	Identification Number	Packing Group
Air, compressed	2.2	UN1002	N/A
Argon, compressed	2.2	UN1006	N/A
Helium, compressed	2.2	UN1046	N/A
Hydrogen, compressed	2.1	UN1049	N/A
Methane, compressed or Natural gas, compressed (<i>with high methane content</i>)	2.1	UN1971	N/A
Nitrogen, compressed	2.2	UN1066	N/A
Oxygen, compressed	2.2	UN1072	N/A

SAFETY CONTROL MEASURES:

a. PACKAGING - Prescribed packaging is a non-DOT specification fiber reinforced plastic (FRP) hoop wrapped (HW) cylinder made in accordance with Pressed Steel Tank's specification presented with their petition on file with the Office of Hazardous Materials Special Permits and Approvals (OHMSPA), and in full compliance with DOT FRP-2 Standard (See Appendix I) Revision 1 dated January 4, 1987 (§ 178.BB) except as follows:

§ 178.BB.-2 Type size and service pressure. Type 3HW cylinder consisting of resin impregnated continuous filament windings in the circumferential direction only over a seamless steel liner made in compliance with 178.BB-6(a); not over 400 pounds water capacity; and service pressure at least 900 PSI but not greater than 5000 PSI. The maximum steel liner sidewall thickness must not exceed 0.500 inches.

§ 178.BB-4 Duties of Inspector.

(a) * * *

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(b) Verify compliance of steel liner with § 178.BB-6(a). Verify compliance of filament and resin system components with the requirements specified in § 178.BB-5.

* * * * *

§ 178.BB-5 Authorized material and identification of material.

(a) Liner material must be as prescribed in § 178.37(b).

* * * * *

§ 178.BB-6 Manufacture.

(a) Liner. Liner without overwrap must be suitable for a marked service pressure of at least 50 percent of the service pressure marked on the composite cylinder, and must be in full compliance with DOT 3AA (§ 178.37) except as follows:

(1) The design is required to be in compliance with § 178.37(a) (2).

* * * * *

(b) Composite Cylinder. The composite cylinder must be fabricated from a steel liner circumferentially wrapped over the entire cylindrical portion with resin impregnated continuous filament windings. Winding pattern to be "hoop" wrap, applied under controlled tension to develop the design composite thickness. After winding is complete, the composite must be cured by a controlled temperature profile, and autofrettaged by pressurizing to not less than 105 and not greater than 115 percent of the prescribed minimum test pressure. No defect is acceptable that is likely to weaken the finished cylinder appreciably.

* * * * *

§ 178.BB-7 Wall thickness.

(a) Minimum thickness of the liner must be at least equal to the minimum design thickness (§ 178.BB-18 (h)) and be such that after autofrettage, the compressive stress in the sidewall of the liner at zero pressure

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will not exceed 50 percent of the minimum yield strength of the steel as determined in § 178.37(k) or 50 percent of the minimum design yield strength shown in § 178.BB-18 (h). The maximum tensile stress of the liner at operating pressure may not exceed 60% of the yield strength.

* * * * *

§ 178.BB-8 Openings.

(a) and (b) * * *

(c) Taper threads when used must comply with one of the following:

(1) American Standard Pipe Thread (NPT) standard must comply with the requirements of Federal Standard H 28/7 (1978).

(2) National Gas Taper Thread (NGT) Standard must comply with the requirements of Federal Standard H 28/7 and H 28/9 (1978).

§ 178.BB-12 Destructive tests.

(a) Applies except change cycling rate to "10 cycles per minute".

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§ 178.BB-13 Acceptable results of tests.

(a) * * *

(b) Physical test. Applies to steel liner only.

(1) Elongation must be at least 20 percent for 2 inch gauge length or at least 10 percent in other cases.

(2) * * *

(3) * * *

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(c) Cycling test.

(1) Each test cylinder must withstand at least 10,000 pressurizations between approximately zero and service pressure, without evidence of distortion or failure.

(d) Burst test.

(1) Burst pressure must be at least 2-1/2 times the service pressure and in no case less than the value necessary to meet the stress criteria of § 178.BB-7(b). Failure must initiate in the cylinder sidewall. Cylinders with marked service pressure not exceeding 2200 psi must remain in one piece. Actual burst pressure must be recorded.

(2) * * *

§ 178.BB-14, § 178.BB-15, and § 178.BB-16

Change "aluminum" to "steel" whenever the word aluminum is found in these paragraphs.

* * * * *

§ 178.BB-18 Design qualification tests.

(a) through (c) * * *

(d) * * * Except change cycling rate to "10 cycles per minute".

(1) and (2) * * *

(3) Not required.

(e) * * *

(1) * * *

(2) See § 178.BB-13(d)(1) of this special permit.

(f) Not required

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b. TESTING - Each cylinder must be reinspected and hydrostatically retested every three years in accordance with § 180.209(a) as prescribed for DOT 3HT cylinders, except that the rejection elastic expansion criteria does not apply, permanent volumetric expansion may not exceed 5 percent of total volumetric expansion at test pressure and retest dates must be imbedded in the epoxy coating in a permanent manner other than stamping. Retest dates may be steel stamped on the shoulder of the top head in accordance with § 178.BB-15(d). Reheat treatment or repair of rejected cylinders not authorized.

c. OPERATIONAL CONTROLS - Cylinders manufactured under this special permit are not authorized for use 15 years from the date of manufacture.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modifications or changes are made to the package and it is offered for transportation in conformance with this special permit and the HMR.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. Each packaging manufactured under the authority of this special permit must be marked with a registration symbol designated by the Office of Hazardous Materials Special Permits and Approvals for a specific manufacturing facility.

e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.

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- f. Use of these cylinders for underwater breathing is not authorized.
 - g. A cylinder which has been subjected to the action of fire may not be returned to service.
 - h. The special filling provisions in § 173.302(c) are not authorized.
 - i. Prototype cylinders to be used solely for design qualification testing may be charged and shipped from the filling facility to the test site, provided each cylinder is in conformance with the provisions of this special permit except for marking (§ 178.BB-15) and design qualification testing (§ 178.BB-18).
 - j. Transportation of Division 2.1 materials (flammable gases) by passenger-carrying aircraft is not authorized.
 - k. Transportation of Division 2.1 materials (flammable gases) are not authorized aboard cargo vessel or cargo aircraft unless specifically authorized in the Hazardous Materials Table (§ 172.101).
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, cargo aircraft only, and passenger-carrying aircraft (see paragraphs 8.j. and 8.k. for restrictions).
10. MODAL REQUIREMENTS:
- a. A current copy of this special permit must be carried aboard each cargo vessel, aircraft or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
 - b. Division 2.1 flammable gases are not authorized to be transported aboard passenger-carrying aircraft.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

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- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- 'The Hazardous Materials Safety and Security Reauthorization Act of 2005' (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term "exemption" to "special permit" and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.



for Dr. Magdy El-Sibaie
Associate Administrator for Hazardous Materials Safety

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Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, Department of Transportation, Washington, D.C. 20590. Attention: PHH-31.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at http://hazmat.dot.gov/sp_app/special_permits/spec_perm_index.htm Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: TG